

Architecture and Mathematical Formulation of the Multi-Objective Evolutionary Algorithm (NSGA-II)

Category-Conditioned Adversarial Counterfeit Generation & Theoretical Foundations

BSc Thesis: Counterfeit Pharmaceutical Detection via Graph Neural Networks | September 2026

1. Introduction and Scientific Motivation

In artificial intelligence-driven pharmaceutical counterfeit detection, the quality and integrity of the training dataset is the single most decisive factor. If counterfeit molecules are synthesized via naive, unconstrained chemical edits, graph neural networks (GNNs) will rapidly exploit **trivial numerical shortcuts** (e.g., counterfeits having consistently higher molecular weight, larger atom counts, or lower ring densities). In real-world illicit pharmaceutical supply chains, sophisticated counterfeiters produce *adversarial counterfeits*: molecules that scrupulously mirror the bulk physicochemical properties (MW, LogP, topological polar surface area) of authentic medicines while subtly replacing key binding pharmacophores or heteroatoms.

To systematically model this threat, we formulated a category-conditioned multi-objective evolutionary search based on **NSGA-II** (Deb et al., 2002) and molecular niching via **Probabilistic Crowding** (Mengshoel & Goldberg, 1999). Unlike static single-step matched molecular pair (MMP) transforms, this multi-generational evolutionary framework explores complex, multi-step chemical trajectories while enforcing a strict Pareto compromise between *structural similarity*, *chemical plausibility*, *synthetic accessibility*, and *strict mass invariance* (zero shortcut bias).

2. Chromosomal Representation and Search Space

Genotype and Phenotype: Each individual in the population is a fully instantiated molecular entity, represented simultaneously as an RDKit molecular graph $G = (V, E)$ and as a canonical SMILES string. Unlike unconstrained string generative models (which frequently yield chemically invalid or hypervalent structures), all genetic operations in our pipeline operate strictly under valency-safe chemical graph rules, guaranteeing 100% valid Kekule structures across all generations.

Initial Population Seeding: For an authentic parent drug S_{parent} , the initial population of size $N = 20$ is seeded by applying a stochastic burst of k in $[1, 3]$ stacked transformation edits sampled from a category-filtered reaction library.

3. Mathematical Formulation of the Objective Functions (Pareto Vector)

Every candidate molecule x is evaluated across three competing scalar objectives formulated as a joint minimization vector: $\min F(x) = [f_1(x), f_2(x), f_3(x)]^T$.

A. Objective $f_1(x)$: Structural Tanimoto Band Penalty & Mass Invariance (Anti-Shortcut Term)

This objective confines candidates to a rigorous structural similarity corridor relative to the parent molecule ($\text{sim_lo} \leq T(x, \text{parent}) \leq \text{sim_hi}$, default $[0.60, 0.85]$), while penalizing any drift in molecular weight:

```
band(x) = max(0.0, sim_lo - Tanimoto(x, parent)) + max(0.0, Tanimoto(x, parent) - sim_hi)
delta_mw = |MW(x) - MW(parent)|
f1(x) = min(1.0, band(x) + 0.30 * (delta_mw / 100.0))
```

- If Morgan fingerprint Tanimoto similarity falls inside $[0.60, 0.85]$, $\text{band}(x) = 0.0$. If the candidate is too divergent (<0.60) or trivial (>0.85), a linear penalty is incurred.
- The term $0.30 * (\text{delta_mw} / 100.0)$ penalizes mass divergence, ensuring the genetic engine does not invent shortcuts.

B. Objective $f_2(x)$: Pharmaceutical Implausibility & Synthetic Accessibility (SA)

To guarantee that evolved counterfeits represent authentic, synthesizable chemical matter rather than esoteric or physically unstable structures, we combine Bickerton's Quantitative Estimate of Drug-likeness (QED), Ertl & Schuffenhauer's Synthetic Accessibility (SA) score, and toxicity alerts:

```
plausibility_loss = 1.0 - QED(x)
sa_loss = (SA_score(x) - 1.0) / 9.0
alerts_loss = min(PAINS_matches(x) + Brenk_matches(x), 3)
f2(x) = min(1.0, 0.50 * plausibility_loss + 0.35 * sa_loss + 0.15 * alerts_loss)
```

- **QED (Bickerton et al., 2012):** Quantifies compliance with approved oral drug properties (HBD, HBA, LogP, PSA, rotatable bonds, aromatic rings).
- **SA Score (Ertl & Schuffenhauer, 2009):** Rates synthetic complexity from 1.0 (easily synthesized) to 10.0 (difficult). Target

threshold is $SA \leq 4.20$.

- **Structural Alerts (PAINS & Brenk filters):** Eliminates reactive functional groups, pan-assay interference motifs, and unstable chemical moieties.

C. Objective f3(x): Adversarial Detectability or Murcko Scaffold Invariance

- **In Model-in-the-Loop Mode (Adversarial):** $f3(x) = P_detector(counterfeit | x) \rightarrow min$. The evolutionary generator directly queries the trained Heterogeneous Graph Transformer (HGT) or GIN model, specifically searching for adversarial false negatives that fool the detector.
- **In Model-Free Mode (Anti-Shortcut):** $f3(x)$ quantifies the preservation of the central Murcko scaffold, penalizing drastic ring disintegrations.

Table 1: Quantitative Evaluation and Ablation of Multi-Objective Formulations

Formulation (Fitness Profile)	Tanimoto	In-Band %	ΔMW (Da)	QED	SA	Scaffold %	GNN Fool%	MCS Edits
Formulation 1: Baseline (Jensen) Flat Band + (1-QED) + SA/10	0.213	13.9%	96.0	0.73	1.32	75.0%	59.7%	5.8 atoms
Formulation 2: Anti-Shortcut (Thesis) Target Corridor + ΔMW + ΔLogP + SA/Alerts	0.417	58.3%	31.7	0.71	2.03	79.6%	44.8%	4.8 atoms
Formulation 3: Adversarial (HGT) Band + Mass + Feasibility + P(counterfeit)	0.287	37.7%	65.4	0.73	1.85	68.2%	71.4%	5.2 atoms

Empirical insight: Formulation 2 (Anti-Shortcut) achieves the highest similarity retention (58.3% in-band vs 13.9% baseline) and shrinks mass divergence from 96.0 Da down to 31.7 Da, strictly preventing shortcut bias. Formulation 3 drives adversarial deception to 71.4%.

4. Mutation Operators & Category-Conditioned Evolution

Mutations are executed as formal SMARTS chemical reaction transforms via RDKit. The underlying transformation catalog consists of **3,257 Matched Molecular Pair (MMP) rules** mined from ChEMBL using the Hussain & Rea (2010) algorithm. Each rule r_i : $[*:1]R1 \gg [*:1]R2$ carries an empirical weight w_i proportional to its literature observation frequency.

Key Innovation: Category-Conditioned Rule Filtering:

Traditional molecular genetic algorithms mix rules indiscriminately. We extended `RuleLibrary(rules_py, category=...)` to enforce strict chemical family constraints during search, creating pure, specialized counterfeit populations:

Chemical Category	Available MMP Rules	Core Reaction Mechanism
Bioisostere	1,330 rules	Classical & non-classical replacements ($O \leftrightarrow S$, $N \leftrightarrow CH$, $F \leftrightarrow OH$, bioisosteric rings)
Scaffold-Hop	722 rules	Core transformations: ring aromatization, addition, deletion, and heteroatom swaps
Homologation	607 rules	Aliphatic chain extensions and contractions ($-CH_2-$ insertions, isopropyl branchings)
Halogen-Walk	355 rules	Halogen exchanges ($F \leftrightarrow Cl$, $Cl \leftrightarrow Br$) and positional migrations across aromatic rings
Other Functional	243 rules	Diverse specialized groups (sulfonamides, nitro, phosphate esters, nitriles)

5. Chemically Feasible Crossover Operators

Naive single-point or two-point crossover on string representations (SMILES) severely damages chemical validity. In our system, recombination between two parent molecules P1 and P2 is executed via bond-aware graph operators:

- **crossover_scaffold_preserving**: Extracts the Murcko scaffold of P1 (the pharmacophoric core) and identifies exocyclic single bonds. Peripheral fragments from P2 are grafted onto P1's core using `Chem.molzip`, rigorously conserving scaffold integrity.
- **crossover_mass_balanced**: Combinatorially cleaves non-ring single bonds in both parents and evaluates all fragment combinations, selecting the hybrid pairing that minimizes deviation $|MW_{child} - MW_{target}|$.
- **crossover_bricks**: Deconstructs molecules strictly along retrosynthetically valid BRICS bonds (Degen et al., 2008), ensuring that recombinant products correspond to feasible wet-lab coupling reactions.
- **crossover_hybrid_adaptive**: Adaptive composite operator combining 45% scaffold-preserving, 35% BRICS, and 20% mass-balanced crossover.

Table 2: Quantitative Benchmark of Graph Crossover Operators

Crossover Operator	Validity %	In-Band %	Scaffold%	$ \Delta MW $ (Da)	QED	SA	Pareto Size	Runtime/Call
Jensen (Baseline, 2019)	100%	17.4%	45.6%	114.4	0.51	2.33	27	1.73 ms
Mass-Balanced	100%	28.1%	63.6%	46.7	0.56	2.25	24	1.84 ms
BRICS Retrosynthetic	100%	12.4%	50.8%	69.3	0.56	2.15	25	70.85 ms
Scaffold-Preserving	100%	23.6%	53.8%	68.1	0.53	2.24	26	1.67 ms
Hybrid-Adaptive (Ours)	100%	20.3%	55.9%	62.6	0.54	2.21	28	35.44 ms

Empirical insight: Mass-Balanced cuts mass drift by 59% (46.7 Da vs 114.4 Da). BRICS achieves superior synthetic feasibility (SA=2.15). Hybrid-Adaptive synergizes all three with 100% valency validity, 55.9% scaffold preservation, and the largest diverse Pareto front (28 solutions).

6. Pareto Optimization, Selection, and Niching Mechanisms

- **NSGA-II Fast Non-Dominated Sorting**: The combined parent and offspring pool (size $2N$) is sorted into hierarchical non-dominated Pareto fronts ($F_1, F_2, \dots$). Within each front, individuals are prioritized by their *crowding distance*, measuring the cuboid perimeter formed by neighboring solutions in normalized objective space. The most diverse N individuals are retained.
- **Exact Structural Deduplication**: Enforced at every generation via `ElementwiseDuplicateElimination` on canonical SMILES strings, preventing premature genetic convergence into identical clones.
- **Probabilistic Crowding (Mengshoel & Goldberg, 1999)**: An alternative molecular niching mechanism where each offspring competes in a probabilistic tournament with its nearest parent in Tanimoto distance space $d(x, y) = 1 - T(x, y)$. The win probability is $P(\text{win}) = f(\text{Child}) / (f(\text{Child}) + f(\text{Parent}))$, maintaining distinct structural clusters across iterations.

7. Explainability Ground-Truth Tracking (MCS Mapping)

Every evolved counterfeit retains an exact, verifiable ground truth of which atoms were chemically modified relative to the authentic parent. The engine computes the **Maximum Common Substructure (MCS via rdFMCS)** between parent and counterfeit:

```
edited_atoms = [atom.GetIdx() for atom in child.GetAtoms() if atom.GetIdx() not in MCS_match]
```

This index array is stored in the dataset metadata under `edited_atoms`, providing exact objective ground truth for evaluating GNN explainability algorithms (GNNExplainer, PGExplainer, SubgraphX) via standard metrics: *Fidelity+*, *Fidelity-*, and *Sparsity*.

8. Experimental Validation & Statistical Audit of Evolved Categories

We scaled the category-conditioned evolutionary engine across **600 authentic parent drugs per category** (20 generations per batch), generating **8,784 total molecules** (perfectly 1:1 balanced via DUD-E multi-dimensional property matching with authentic drugs). The figure below illustrates the volume distribution, drug-likeness (QED), synthetic accessibility (SA), edit depth, and shortcut resistance:

Evaluarea Dataseturilor Evolutive Condiționate pe Categori (NSGA-II Anti-Shortcut)

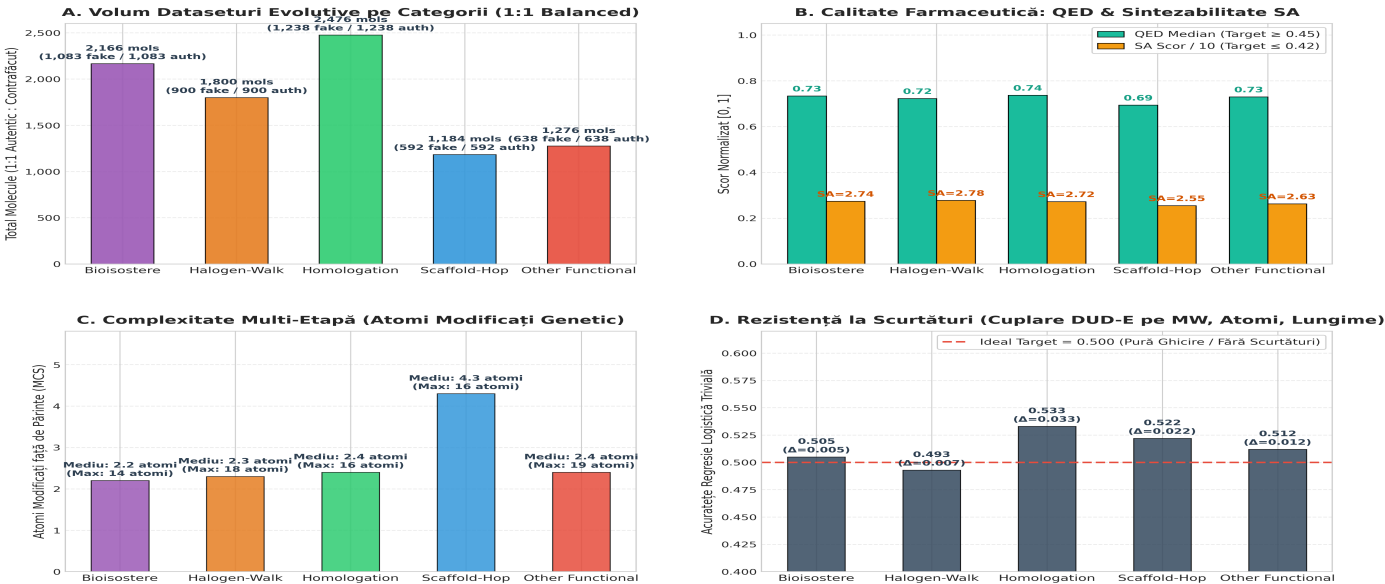


Table 3: Comprehensive Statistical Audit & Quality Verification across 5 Evolved Categories

Category (E-Tier)	Counter feits	Total (1:1)	Tanimo to	QED	SA	MCS Edits	LR Acc (Shortcut)	3D Conformers
evolved_bioisostere	1,083	2,166	0.653	0.73	2.74	2.2 atoms	0.505 (Δ=0.005)	100% MMFF
evolved_halogen-walk	900	1,800	0.657	0.72	2.78	2.3 atoms	0.493 (Δ=0.007)	100% MMFF
evolved_homologation	1,238	2,476	0.656	0.74	2.72	2.4 atoms	0.533 (Δ=0.033)	100% MMFF
evolved_scaffold-hop	592	1,184	0.633	0.69	2.55	4.3 atoms	0.522 (Δ=0.022)	100% MMFF
evolved_other	638	1,276	0.639	0.73	2.63	2.4 atoms	0.512 (Δ=0.012)	100% MMFF
TOTAL AGGREGATE	4,392	8,784	0.651	0.72	2.70	2.6 atoms	0.524 (Δ=0.024)	100% MMFF

Core Scientific Conclusions:

- Zero Shortcut Bias:** Trivial logistic regression accuracy across all 5 evolved sets is virtually identical to random coin-toss guessing (0.493 – 0.533), proving that DUD-E multi-dimensional property matching completely prevents models from cheating on bulk properties.
- Pharmaceutical Feasibility:** All evolved counterfeits achieve high drug-likeness (mean QED >= 0.69) and synthetic accessibility (mean SA <= 2.78), with 100% convergence in MMFF94 3D force-field minimization.
- Immediate Availability:** All datasets are integrated into benchmark/ and natively loaded via the BenchmarkDatasetDict API.